LOADER , WEAPONS, AIR LAUNCHED, A/S32K-1D S32K-1D D

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers an air launched weapons loader developed as part of the U.S. Marine Corps Expeditionary Airfield Program. The loader will be designated as Model A/S32K-1D and referred to as the second generation loader or the "loader." The following modifications are present in the loader:

a. A Diesel engine of increased horsepower.

b. A new hydrostatic propulsion system.

c. Improved hydraulic system.

d. A new power steering system.

e. Improved hydraulic fluid cooling capability.

f. Improved disc type brakes.

g. Improved actuating controls for lift system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form

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SPECIFICATIONS

FEDERAL

TT-C-494	Coating Compound, Bituminous, Solvent Type, Acid Resistant
VV-F-800	Fuel Oil, Diesel

MILITARY

MIL-V-173	Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment)
MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-G-3855	Grease Gun, Hand Operated, Lever, Push and Screw Type
MIL-G-5514	Gland Design, Packings, Hydraulic, General Requirements for
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-Q-9858	Quality Program Requirements
MIL-W-52574	Welding Process and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels
MIL-M-81807	Marking of Aircraft Ground Support Equipment with Retro-Reflective Materials, Process for
MIL-B-83769/6	Battery, Storage, Lead Acid

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Services
FED-STD-595	Color

MILITARY

MIL-STD-28 1	Automobiles, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of

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STANDARDS (Continued)

MILITARY (Continued)

MIL-STD-461 1	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic INterference
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-808	Finish, Protective and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program
MIL-STD-1474	Noise Limits for Army Material

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 1378AS100	Weapons Loader, Air Launched, 4500 Pounds Capacity
137AS225	Painting Procedures

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein,

UNITED STATES DEPARTMENT OF AGRICULTURE

USDA - Forest Service Standard 5100-1

(Application for copies should be addressed to San Dimas Equipment Development Center, 444 E. Bonita Avenue, San Dimas, CA 91773.)

(Copies of specifications. standards, handbooks, drawings, and publications required by manufacturers in connection with specific equisitor functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles , CA 90013.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers , Inc. , 400 Commonwealth Drive, Warrendale, PA 15086.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

301 First article. When specified, a loader shall be subjected to first article inspection specified herein (see 4.4 and 6.3).

3.2 Material. The material used shall be as specified herein. However, material not specified herein shall be of a quality suitable for the purpose intended and to meet the requirements of this specification. The material shall be free of all defects and imperfections that might affect the serviceability of the finished product.

3.2.1 Metals.

3.2.1.1 Dissimilar metals. Dissimilar metals (nonpermissible galvanic couples) as defined in MIL-STD-889, Type II shall not be used in contact with each other. Where contact is unavoidable, dissimilar metals shall be insulated as specified in MIL-STD-889, section 5.

3.2.2 Nonmetallic materials.

3.2.2.1 Tire material. Tires shall not exceed eight calendar quarters in age from date of cure of the rubber to date of acceptance of the loader by the Government.

3.3 Parts.

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2 .3.1 Standard parts. Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their military part numbers. All parts of the same part number shall be interchangeable as applied within the system.

3.3.2 Nonstandard Parts. Use of nonstandard parts requires approval by the acquiring activity. Request for approval for the use of nonstandard parts shall be submitted in accordance with MIL-STD-965.

3.3.3 Commercial hardware parts. Commercial utility hardware items such as bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by standard parts (specification sheet, MS and AN) without alteration and providing the corresponding standard part numbers are referenced in the parts list and on the contractor's drawing.

3.4 Interchangeability. All parts shall be manufactured to standards that will permit replacement or adjustment without modification of parts.

3.5 Finish.

3.5.1 Corrosion resistant metal parts. All protected metal parts and surfaces that are constructed of corrosion resistant material and on which a protective coating would be detrimental to proper functioning. need not be given protective coatings. All exposed metal parts and surfaces, and all protected metal parts on which a protective coating will not be detrimental to proper functioning shall be treated for conformance to the requirements of this specification and drawings listed on DL 1378AS100.

3.5.1.1 Protective coating. All exposed parts and surfaces of the equipment and its components shall be given protective coatings as specified herein and on DL 1378AS100.

3.5.2 Surfaces. Ail surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, burrs, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled, unless sharpness is required to perform necessary functions.

3*5.3 Treatment and painting. The loader and equipment shall be treated and finished in accordance with drawing 1378AS225. Surfaces of components and assemblies of the loader not painted with a finish coat , shall be cleaned and prepared in accordance with standard commercial practice. Plastics, rubber, or fabric hoses and drive belts shall net be painted, Engine accessories (i.e., alternator, starter), instruments , and other such end items that are plated or painted by the manufacturer shall not be painted by the supplier and shall be adequately protected or masked from overspray.

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3.6 Fungus and moisture resistance. When specified in the contract (see 6.2.1h), the electrical circuitry, including all components and connections, except as specified below shall be protected from. fungus growth and moisture by an overall treatment of varnish. Varnish shall conform to MIL-V-173, Composition I or II, as applicable (see 3.6.1), with one percent copper 8 quinolinolate (by weight) based on the nonvolatile content of the varnish. Components that do not req'dire protection are:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed,

b. Components or circuit elements whose functions will be adversely affected by the varnish coating,

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 nil to component or element surfaces. The surfaces shall be previously cleaned and prepared so that the surfaces are free of all foreign matter that would interfere with the adherence or function of the varnish,

3.6.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition 1, When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall ceritfy that the Composition II material complies with Rule 66, Air Pollution Control District County of Los Angeles, CA.

3.7 Design and construction,

3.7.1 Design. The design of the loader shall conform to the requirements specified herein and in accordance with DL 1378AS100. The loader shall be capable of performing all the operations specified herein.

3.7.2 Construction. The loader shall be a heavy-duty unit driven by a diesel engine. It shall have pneumatic tires, steerable rear wheels, and two wheel hydrostatic drive. The loader shall be complete with necessary operating accessories (rollers, forks, belts, etc.). There shall be no exposed belts, clamps, gages, fittings, hoses, lifting pads or other appendages on the loader that can be hooked while the loader is working in a confined space.

3.7.2.1 Dimensions. The loader dimensions shall be as follows:

a. Length - 202 inches.

b. Width - 72 inches (maximum).

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c. Height - 44 inches (maximum).

d. Weight - 6600 pounds (maximum).

3,7.2.2 Fastening devices, All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.7.2.3 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.7.2.4 Oil servicing provisions. Oil filler caps shall be painted gloss yellow (color No. 13538) in accordance with FED-STD-595. The oil to be used and the tank capacity shall be stenciled adjacent to the filler opening. The stenciling color shall be in accordance with MIL-STD-808 . The height of the stenciling shall be two inches.

3.7.2.5 Lubrication, All oil holes, grease fittings, and filler caps shall be accessible. Fittings shall be accessible to a hand operated grease gun, conforming to MIL-C-3859, with a 10-inch flexible extension.. A lubrication list in accordance with MIL-STD-838 shall be provided with the loader. The loader shall operate as specified herein when lubricated with military lubricants.

3.7.2.6 Hydraulic packing installation. All hydraulic packings shall be installed in accordance with MIL-G-5514.

3.7.3 Subsystems.

3.7.3*1 Exhaust system, A spark arrestor conforming to the material and performance requirements of U.S. Department of Agriculture Forest Service Standard 5100-1, with the exception of an increase in arrestor efficiency to a 95 percent mean between small and large carbon particles for each tested flow rate, shall be furnished on the loader, The exhaust system shall be installed underneath the loader and the tailpipe located so as to prevent the accumulation of carbon monoxide at the operator^{s} location.

3.7.3.2 Electrical system. The loader shall have a nominal 24-volt electric system. Multiple conductor routing shall be protected by flexible tubing or electrical tape. Grommets shall be provided whenever wiring passes through bulkheads, partitions, or structural members, and the wiring shall be fastened to prevent chafing and abrasion, All wiring shall be color or number coded and be adequately sized, and all circuits shall have suitable overload protection. Fuses and circuit breakers shall be grouped in convenient locations and suitably marked for size and function,

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3 .7.3.2.1 Battery. Lead-acid batteries supplied for the loader electrical system shall conform to MIL-B-83769/6, part number M83769/6-1.

3. 7.3.2.1.1 Battery mounting. Battery mounting shall be in an accessible enclosed compartment. The battery location shall provide for removal of battery without chafing or other damage to the battery. Batteries shall by located so they can be cleaned and serviced without removing any component. Battery supports, holdowns, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with a coating conforming to TT-C-494, Type II. The battery compartment shall have provision for drainage and provision for gas venting at or near the top of the compartment. Cover and positioning shall protect against short circuiting. The nonground cable shall be protected by rubber grommets or insulated passages at its entry to the battery compartment.

3.7.3.3 Fuel system. The fuel system shall consist of a fuel tank, a fuel filter, a fuel pump, connecting fuel lines and an cil bath air cleaner. Use only diesel fuel conforming to VV-F-800 or JP-5 turbine fuel conforming to MIL-T-5624.

3. 7.3.3.1 Fuel tank. The fuel filler cap shall be painted lusterless red, color No. 31136 of FED-STD-595. The fuel type and tank capacity shall be stenciled o the loader adjacent to the fuel filler opening. Stenciling color shall be in accordance with MIL-STD-808. Stenciling height shall be two inches.

3.7.3,3.2 Fuel filter. A fuel filter of adequate capacity shall be provided in the fuel system.

3.7.3.4 Brakes.

3.7.3.4.1 Service brakes. Loaders shall be provided with hydraulic powered brakes for both rear and front wheels. The brakes shall be capable of holding the loader, with 4500 pound weapon or store, on a 20 percent grade. The brakes shall also be capable of locking all four wheels while traveling at maximum speed in the fully loaded condition (see 4.8.2.6).

3.7.3.4.2 Parking brakes, The loader shall be provided with a parking brake, easily accessible to the operator, capable of holding the loader with a 4500 pound weapon or store, on a 20 percent grade (see 4.8.2.7).

3.7.3.4.3 Accumulator, The service brake hydraulic system shall be provided with an accumulator which is charged to capacity by an engine driven pump. The accumulator shall be capable of being charged while the engine is operating at 1500 rpm. It shall also be capable of delivering not less than five emergency stops in the event of an engine failure (see 4.8.2.8).

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3. 7.3.5 Steering. Steering of the loader shall be accomplished by a power steering mechanism. The steering gear ratio shall insure positive control of the loader while traveling at maximum speeds. The rear wheel steering shall be capable of turning 30 in either direction fore and aft of the centerline of the loader (see 4.8.2.9) .

3.7.3.6 Front wheels. The distance between the front wheels, mounted on the forward side frames, shall be individually, infinitely adjustable, and hydraulically actuated from 72 to 144 inches (see 4.8.2.10).

3.7.4 Lifting mechanism and manipulating head.

3.7.4.1 Lifting mechanism. The hydraulic lifting mechanism shall be capable of lifting a 4500 pound weapon, store or any combination thereof in any given attitude, from two inches above ground to a minimum of 89 inches or 105 inches above the ground depending on the normal or inverted position of the forks. The lifting mechanism shall be equipped with a "fail safe" device to prevent the load from creeping or falling while in any attitude, with the engine in the off position (see 4.8.2.11).

3.7,4. 1.1 Lifting rate. The hydraulic lifting mechanism shall have a lifting rate, up or down, loaded or empty, of not less than 30 feet per minute throughout the entire lift range of 2 to 89 inches (see 4.8.2.11).

3.7.4. 1.2 Incremental control. The lifting mechanism shall provide fully controlled lift that can be controlled in increments of l/8-inch or less throughout the entire lift cycle (see 4.8.2.11).

3.7.4. 1.3 Azimuth mechanism. The lifting boom shall be provided with an azimuth mechanism that positions the lifted weapon or store by horizontally moving the boom 20 degrees to either side of center and shall be made to hydraulically control this movement in increments of l/8-inch or less (see 4.8.2.11.1).

3.7.4. 1.4 Overload. The lifting boom shall be capable of holding a static overload of 13,500 pounds for not less than three minutes without damage to any of the components to prevent the loader from performing its operational service life expectancy (see 4,8.2,11.2).

3.7.4, 1.5 Lifting time for auxiliary hand pump system. In case of a prime power failure the lifting mechanism shall be provided with an auxiliary system that has a lift time, under load, of not more than five minutes for the entire lift range of 2 to 89 inches (see 4.8.2.11.3),

3.7.4.2 Manipulator head. The lifting boom shall be provided with a manipulator head for restraining and positioning the weapon or store. The head shall be equipped in such a manner as to have five separate motions (see 4.8.2.12).

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a. Yaw. The manipulator head shall impart to the weapon or store a yaw of 90 degrees both right and left.

b. Pitch (tilt) The manipulator head shall impart to the weapon or store a tilt movement of 15 degrees clockwise or counterclockwise.

c. Lateral. The manipulator head shall impart to the ~'capon or store a total lateral movement of 4 inches.

c?. Longitudinal. The manipulator head shall impart to the weapon or store a total longitudinal movement of 3 inches.

e. Roll. The manipulator head shall impart to the ~'capon or store, a total roll movement of 7 degrees up, and 13 degrees down from the level position, with load on rollers or adapter of forks, 360 ,

3.7.4.3 Forks. With the lifting boom in its lowest position, th fork lift tines shall be easily removable form the manipulator head and capable of being positioned as follows (see 4.8.2.13).

a. Top of forks (facing forward) 2 inches above surface on which loader sits (the tires forward before fork support).

b. Top of forks (facing forward) 16 inches above surface on which loader sits (tines forward above fork support).

c. Top of forks (facing aft) 16 inches above surface on which loader sits (tines aft above fork support).

3.7.4.3.1 : Fork tine centers. The manipulator head of the loader shall be drilled to position the fork tines at distances of 14, 16, 18, 24, 26 and 28 inches between fork centers.

3.7.4.3.2 Rollers. The fork tines shall be provided with spaced holes for positioning the rollers to accommodate various diameter k'capons.

3.8 Performance requirements.

3.8.1 Mobility . Loaders supplied under this specification shall be capable of meeting, without damage, the mobility requirements listed below. In the event of conflict between this specification and the drawings listed on DL 1378AS100, the requirements stated on the drawings shall apply.

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3.8.1.1 Speed. The loader, while running on a dry, clean, smooth, level surface, with a 4500 pound weapon or store, shall be capable of attaining a maximum forward speed of not less than 8 miles per hour and not more than 10 miles per hour (see 4.8.2.1).

3.8.1.2 Climbing capability, The loader shall be capable of climbing a 20 percent grade without a load and climbing a 10 percent grade with a 4500 pound load or store without stalling (see 4.6.2.2).

3.8.1.3 Side slope stability. The loader with a 2000 pound weapon or store on the forks or cradle, shall be capable of negotiating a 10 percent side slope with the boom and manipulator in any position (see 4.8.2.3).

3.8.1.4 Turning radius. The loader shall be capable of turning a 20-foot radius. The loader shall turn in one motion without backing (see 4.8.2.4).

3.8.1.5 Road test. The loader shall be capable of being driven at speeds varying between 0 and 10 miles per hour for a distance of 50 miles without a load and 50 miles with a 4500 pound load or store for a total of 100 miles without evidence of damage or construction deficiencies (see 4.8.2,5],

3.8.2 Cycle. The loader shall be capable of meeting the operational requirements during its service life without damage to any components when subjected to 250 cycles of the cycle test (see 4.8.3).

3.8.3 Noise level. The noise level of the loader shall be 85db(A) or less at the operator's station (see 4.8.4).

3.8.4 Electromagnetic radiation. Unless otherwise specified the loader shall be equipped for the suppression of radio interference to meet the requirements of MIL-STD-461 for Class IIIC equipment (see 4.8.14).

3.8.5 Environmental. The loader shall be capable of operating without damage when subjected to the following environmental conditions in MIL-STD-810.

3.8.5.1 High temperature, The loader shall be capable of operation under high temperature conditions (see 4,8.6.1).

3.8.5.2 Low temperature. The loader shall be capable of operation under low temperature conditions (see 4.8.6.2).

3.8.5.3 Rain. The loader shall be capable of operation under rain conditions (see 4.8.6.3).

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3.8.5.4 Humidity. The loader shall be capable of operation under humid conditions when tested in accordance with 4,6.6,4,

3.$.5.5 salt fog. The loader shall be capable of operation under salt fog conditions when tested in accordance with 4.8.6.5.

3.8.5.6 Fungus resistance. Tile loader shall be capable of operation when subjected to fungus resistance conditions when tested in accordance with 4.8.6.6.

3.8.5.7 Sand and dust. The loader shall be capable of operation under sand and dust conditions when tested in accordance with 4.8.6.7.

3*9 Servicing and adjusting. Prior to the delivery of the loader, the contractor shall service and adjust each vehicle for operational use, including the following: focusing of lights; adjustment of the engine, hydraulic, electrical and brake systems; alignment of rear wheels: inflation of all tires; and complete lubrication of chassis, engine, and rear running gear, with grades of lubricant recommended for the ambient air temperature at the delivery point.

3.10 Markings.

3.10.1 instruction plates. Instruction plates shall conform to MIL-P-514 Type II, Composition. C. Instruction plates shall be suitably located, shall contain clear concise instructions for the operation of all controls, and shall describe the procedure to be followed in servicing the loader,

3.10.2 Stenciling. The capacity of each unit shall be stenciled on both sides of the boom in such a manner as to be readily discernible. The prescribed tire pressure shall be stenciled on the leader frame above each tire using block or stencil type letters not more than one inch high,

3.10.3 Retro-reflective markings. The loader shall be marked with retro-reflective material in accordance with MIL-M-81807. The tape shall conform to 3M RRT “92-90” white or equivalent.

3.11 Workmanship. The workmanship displayed in the fabrication and assembly of the loader shall be such as to assure, with design limitations, the ability of the loader to meet the performance requirements under all operating and environments-~ conditions specified herein. In addition, workmanship standards shall apply to the following conditions.

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3.11.1 Castings and forgings. All parts, components, and assemblies of the loader which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, burrs, scale, and flux. Rework shall be limited to procedures that do not reduce mechanical properties or affect function. Castings with fastener holes shall provide for correct seating of fasteners such as bolt, lockwasher and nut assemblies and capscrews.

3.11.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall 'be square and true. Flame cutting using tips suitable for the thickness of the steel may be employed instead of shearing or sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges and corners are required or where they are not detrimental to safety.

3.11.3 Welding. The surfaces of parts to be welded shall be free of rust, scale, paint, grease and mill 11 scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the loader to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be in close contact and shall not be separated by more than 3/16-inch unless appropriate bridging techniques are used. The welding process and the welding procedure used in fabrication of the loader shall be in accordance with MILL-W-52574 .

3.11.4 Bolt connections. Bolt holes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall-have full thread engagement.

3.11.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and on DL 1378AS100, and to the standards of the Materials Handling Equipment Industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government, The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements .

4.1.1 Quality program. The contractor shall provide and maintain an quality program in accordance with MIL-Q-9858.

4.1.1.: Test equipment and inspection facilities, The contractor shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Component and material inspection. The contractor shall be responsible for assuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspections. The inspections specified herein. are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Inspection comparison (see 4.6).

4.3 Inspection of packaging. The sampling and inspection of the preservation, packaging, packing and container markings shall be in accordance with the requirements of section 5,

4.4 First article inspection, The loader shall be subjected to the tests marked "X" listed in the first article test column of table I and examined for the defects marked "X" listed in the first article test column of table 11, Acceptance of the first article sample loader shall not exclude the remaining loaders from the quality conformance inspection and acceptance provisions specified in 4,5,

4.4.1 First article inspection conditions. Before first article inspection testing, the following shall be performed:

a. Examine the loader for defects.

b. Service the loader with oils, grease, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted,

c, Precondition the loader in accordance with MIL-STD-781, paragraph 5.1,9. Precondition duration and procedures shall be included as part of the schedule of maintenance d. below.

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d. Furnish a schedule of maintenance to be followed during all testing of the loader. Provide a list of special tools required to perform this maintenance.

4.4.2 First article sample. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit one loader for inspection to determine conformance with this specification (see 6.6.2) and DL 1378AS100. This unit shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of a first article sample on the new contract may be waived at the discretion of the Government (see 6,3.1). Approval of the first article sample, or the waiving of the first article inspection is not a waiver of the requirement of submitting to the quality conformance inspection. Upon completion of the first article inspection, all of the applicable inspection and test reports, and when applicable recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant Government activity (see 6.2). The approved unit shall be returned to the manufacturer for use in monitoring production. It shall be refurbished and delivered as the last unit under the contract.

4.4.3 First article inspection report. A first article inspection report shall be prepared in accordance with MIL-STD-831 (see 6.2.2).

4.5 Quality conformance inspection. The loader shall be subjected to the tests marked "X" listed in the quality conformance test column of table I. Failure of any test shall be cause for rejection. After successful completion of all the quality conformance tests, the loader shall be examined for the defects marked "X" listed in the quality conformance test column of table II. Presence of one or more defects shall be cause for rejection.

4.5.1 Quality conformance inspection report. A quality conformance inspection report shall be prepared in accordance with MIL-STD-831 (see 6.2.2).

4.6 Inspection comparison. The Government may select the loaders at any time during the contract production period and subject the loaders to the examination specified in 4.5 and the first article test column of table II and to tests specified in 4.5 and in the first article test column of table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at random, from those which have been accepted by the Government but will not include the previously inspected first produced loader. In addition to any test specified as part of the inspection comparisons the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure to such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.6.1 Inspection failure. Failure of an inspection comparison to meet any requirement specified in 4,8 shall be cause for rejection of the inspection comparison loader and shall be cause for refusal by the Government to continue acceptance of production loaders until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on loaders previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.7 Test conditions. Unless otherwise specified herein, all inspection tests shall be conducted at the ambient temperature and climatic conditions existing at the place of the test. Only that maintenance established by the contractor and submitted as a maintenance schedule prier to commencement of the tests shall be performed during the tests.

4.7.1 Test loads. Weapons or stores weighing 2000, 4500 and 13,500 pounds respectively shall be used for specific tests requiring these test loads.

4.8 inspection methods.

4.8.1 Visual examination. Each loader shall be visually examined to determine conformance to the requirements specified herein and in DL 1378AS100 with respect to material, parts, finish, cleanliness, service and adjustment, electrical and hydraulic connections, interchangeability, physical dimensions, markings and workmanship.

4.8.2 Mobility characteristics.

4.8.2.1 Speed. The loader shall be tested in accordance with Test Method No. 1 of Appendix A.

4,8.2.2 Climbing capability. The loader shall climb a 20 percent grade, with no load, without stalling and climb a 10 percent grade, with a 4500 pound load, without stalling.

4,8.2.3 Side slope stability. The loader with the side frames fully extended and with a 2000pound weapon or store load on the forks shall negotiate a 10 percent (5° 43′) slide slope with the boom and manipulator head in any position.

4.8.2.4 Turning radius. The loader shall be tested in accordance with Test Method No. 2 of Appendix A,

4.8.2,5 Road test. The loader shall be driven over a total distance of 100 miles in two 50 mile tests. The loader shall be driven over a paved road for 50 miles, unloaded at varying speeds between 0 and 10 miles per hour, At the conclusion of this test the loader shall then

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be driven over a paved road for 50 miles with a 4500 pound weapon or store at varying speed between 0 and 10 miles per hour. During the road test the vehicle shall be stopped using the service brakes at the end of each 1/3 mile. At the conclusion of each test the loader shall show no evidence of damage or construction deficiencies.

4.8.2.6 Service brakes. The loader, with a 4500 pound weapon or store, shall be placed on a 20 percent (11° 19') grade, nose down, and the service brakes applied. There shall be no forward movement within 15 minutes. The brakes shall lock all four wheels when applied while traveling at maximum speed under full load conditions.

4.8.2.7 Parking brakes. The loader, with a 4500 pound weapon or store, shall be placed on a 20 percent (11° 19') grade, nose down. with the parking brake applied, there shall be no forward movement of the loader.

4.8.2.8 Accumulator . The service brake hydraulic system shall demonstrate its ability to charge the accumulator to capacity while tune engine is running at 1500 rpm. With the engine off, tune brake system shall demonstrate the ability to deliver not less than five emergency stops.

4.8.2.9 Steering. The rear wheel steering mechanism shall demonstrate the ability to turn thirty degrees in either direction off the centerline.

4.8.2.10 Front wheels. The front wheel side frames shall demonstrate the ability to be hydraulically actuated and infinitely adjustable from 72 inches to 144 inches (outside wheel track width).

4.8.2.11 Lifting mechanism. The lifting mechanism shall demonstrate the ability to lift a 4500 pound weapon or store at a rate of thirty feet per minute, using the rear controls and the engine running at 2400 rpm, from 2 to 89 inches above the ground. It shall be demonstrated that control increments of 1/8 inch or less can be provided through the lift range. The "fail safe" device shall demonstrate the ability to prevent any downward movement when the engine is off and the pressure line is disconnected from the lift cylinder.

4.8.2. 11.1 Azimuth mechanism. The azimuth mechanism shall demonstrate its ability to move a 4500 pound weapon or store horizontally 20 degrees to either side of center and in 1/8 inch increments or less.

4.8.2.11.2 Overload test. The loader shall be subjected to a load of 13,500 pounds held 18 inches off the ground for three minutes. At the conclusion of the test there shall be no evidence of yield in any of the components that could in any manner prevent the loader from meeting operational service life requirements.

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4.8.2. 11.3 Lifting time for auxiliarv hand pump system. The auxiliary hand pump -system shall demonstrate the ability to lift a 4500 pound weapon or store from 2 to 89 inches in five minutes or less with the engine off.

4.8.2.12 Manipulator head. The loader with a 4500 pound weapon or store in place shall demostrate the ability to move the weapon or store longitudinal centerline in the following manner:

a. Yaw - Yaw 90 degrees both right and left.

b. Pitch (tilt) - Tilt 15 degrees both up and down.

c. Lateral - Move laterally 4 inches.

d. Longitudinal - Move longitudinally 3 inches.

e. Roll - Roll 17 degrees up and 13 degrees down.

4.8.2.13 Forks. When the lifting boom is in its lowest position, it shall be demonstrated that the fork tines car, be easily removed from the manipulator head.

4.8.3 Cycle. The loader shall be subjected to a cycle test in accordance with the procedures of 4.8.2.11, 4.8.2. 11.1 and 4.8.2.12 for 250 cycles for each test procedure. At the end of these tests the loader shall show no evidence of yield in any component that could in any manner prevent the loader from meeting the operational requirements.

4.8.4 Noise limits. The loader shall be subjected to duty cycle tests in lieu of the steady state test of MIL-STD-1474, The continuous noise level (Lcq*) shall be measured at the operator's station in accordance with MIL-STD-1474. Accumulative noise exposure measurement shall require a minimum of one hour of unloaded continuous operation on the test course. Throughout the test the loader shall meet the noise level requirements.

4.8.5 Electromagnetic radiation. The loader shall be tested in accordance with MIL-STD-461 and shall meet the requirements of 3.8.4. in addition, the electromagnetic radiation report shall include results of all testing and a list of components used to reduce the electromagnetic radiation.

4.8.6 Environmental tests.

4.8.6.1 High temperature, The high temperature test shall be conducted in accordance with Test Method No. 3 of Appendix A.

4.8.6.2 Low temperature, The low temperature test shall be conducted in accordance with Test Method No. 4 of Appendix A.

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4.8.6.3 Rain. The rain test shall be conducted in accordance with MIL-STD-810, Method 506. 1, Procedure I.

4.8.6.4 Humidity, The humidity test shall be conducted in accordance with Test Method No. 5 of Appendix A, except that the humidity cycles shall not exceed 5.

4.8.6.5 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810, Method 509.1, Procedure I.

4.8.6.6 Fungus resistance. The fungus resistance test shall be conducted in accordance with MIL-STD-810, Method 508.1, Procedure I.

4.8.6.7 Sand and dust, The sand and dust test shall be conducted in accordance with MIL-STD-810, Method 510.1, Procedure 1.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packing shall conform. to MIL-STD-2 81, level A (mobile) or level C (mobile.) and as specified In the contract The dash panel shall be protected against transportation damage.

6 NOTES

6.1 Intended use. The loader is designed specifically for transporting, loading, and unloading weapons and stores weighing up to 4500 pounds to and from the wing racks and centerline pylons of U.S. Marine Corps Tactical Aircraft. The loader will be operated on prepared surfaces such as concrete and asphalt.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Level of preservation and packaging required (see 5.1).

c. Special marking, if required.

d. First article inspection, if other than specified (see 6.3.1).

e. Where first article inspection shall be conducted (see 6.3. la).

f. Where first article inspection report shall be sent.

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g. Time frame required for submission of first produced loader.

h. Whether fungus and moisture resistance is required (see 3.6).

i. Whether electromagnetic radiation emission protection in accordance with MIL-STD-461 is required (see 3.8.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delievered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
(a) 4.1.1	Inspection system program plan	DI-R-4803	---
(b) 4.4.3	inspection and test reports	DI-T-5329	---
(c) 4.5.1	Inspection and test reports	DI-T-5329	---
(a) 6.5	Failure report	UDI-R-21141	---

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article inspection. Contracts or orders should specify the following provisions for the first article inspection.

6.3.1 Whether first article inspection is required. When a contractor is in continuous production of the loader from contract to contract, consideration should be given to waive the first article inspection If first article inspection is required, indicate:

a. Whether the first article inspection is to be conducted at the contractor's plant, Government or other facilities.

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b. That the approval of the first article sample, or the waiving of the first article inspection} shall not relieve the contractors of their obligation to fulfill all other requirements of specification, drawing list, and contract.

6.4 First produced loader. Any changes or deviations of production loaders from the approved first produced loader during production will be subject to the approval of the contracting officer, Approval of the first produced loader will not relieve the contractors of their obligation to furnish loaders conforming to the specification.

6.5 Incident report. When the contractor conducts the tests specified herein, a preliminary written report shall be furnished to the contracting officer within 24 hours of any incident of equipment malfunction or failure during the testing. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected and description of incident. A final report shall be submitted within 15 days.

6.6 Definitions.

6.6.1 Failure. A failure is classified as any condition that will preclude acceptance of the equipment for compliance with test/ Inspection requirements specified herein, Failure actions of MIL-STD-781, paragraph 5.6, shall apply.

Preparing activity:
Navy - AS
(Project No, 1740-N142)

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TABLE I. Test schedule.

First Article Test	Quality Conformance Test	Test Method	Requirement Paragraph	Test Paragraph
X	X	Visual examination	3.2, 3.3, 3.5, 3.7.2, 3.9, 3.10, 3.11	4.8.1
X	X	Speed	3.8.1.1	4.8.2.1
X	X	Climbing capability	3.8.1.2	4.8.2.2
X		Slide slope stability	3.8.1.3	4.8.2.3
X	X	Turning radius	3.8.1.4	4.8.2.4
X	X	Road test	3.8.1.5	4.8.2.5
X		Exhaust system	3.7.3.1	4.8.3
X	X	Service brakes	3.7.3.4.1	4.8.2.6
X	X	Parking brakes	3.7.3.4.2	4.8.2.7
X		Accumulator	3.7.3.4.3	4.8.2.8
X	X	Steering	3.7.3.5	4.8.2.9
X		Front wheels	3.7.3.6	4.8.2.10
X	X	Lifting mechanism	3.7.4.1	4.8.2.11
X	X	Lifting rate	3.7.4.1.1	4.8.2.11
X		Incremental control	3.7.4.1.2	4.8.2.11
X		Azimuth mechanism	3.7.4.1.3	4.8.2.11.1
X	X	Overload	3.7.4.1.4	4.8.2.11.2
X	X	Lifting time for auxiliary hand pump system	3.7.4.1.5	4.8.2.11.3
X		Manipulator head yaw	3.7.4.2	4.8.2.12

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TABLE I. Test schedule (Continued)

First Article Test	Quality Conformance Test	Test Method	Requirement Paragraph	Test Paragraph
X		Pitch	3.7.4.2	4.8.2.12
X		Lateral	3.7.4.2	4.8.2.12
X		Longitudinal	3.7.4.2	4.8.2.12
X		Roll	3.7.4.2	4.8.2.12
X		Forks, normal position	3.7.4.3	4.8.2.13
X		Forks, inverted facing forward	3.7.4.3	4.8.2.13
X		Forks, inverted facing backward	3.7.4.3	4.8.2.13
X		Fork tine centers	3.7.4.3.1	4.8.2.13
X		Rollers	3.7.4.3.2	4.8.2.13
X	X	Cycle	3.8.2	4.8.3
X		Noise level	3.8.3	4.8.4
X		Electromagnetic radiation	3.8.4	4.8.5
X		High temperature	3.8.5.1	4.8.6.1
X		Low temperature	3.8.5.2	4.8.6.2
X		Rain	3.8.5.3	4.8.6.3
X		Humidity	3.8.5.4	4.8.6.4
X		Salt fog	3.8.5.5	4.8.6.5
X		Fungus resistance	3.8.5.6	4.8.6.6
X		Sand and dust	3.8.5.7	4.8.6.7

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TABLE II. Examination schedule.

First Article Test	Quality Conformance Test	Defect	Drawing Number	Requirement Paragraph
X	-	101 Materials not as specified.	-	3.2
X	-	102 Lubrication fittings not as specified.	-	3.7.2.5
X	-	103 Screw threads not as specified.	-	3.7.2.3
X	X	104 Dipsticks not as specified.	1378AS130	
X	-	105 Alternator and regulator not as specified.	1378AS160	
X	-	106 Starting system not as specified.	1378AS160	
X	-	107 Fuel system not as specified.	1378AS107	
X	-	108 Cooling system not as specified.	1378AS109	
X	X	109 Governor speed not as specified.	1378AS130	
X	-	110 Exhaust system not as specified.	1378AS109	
X	-	111 Fungus and moisture resistance not as specified.	-	3.6
X	-	112 Hydraulic Oil filters not as specified.	1378AS239	
X	-	113 Hydrostatic transmission not as specified.	1378AS106	
X	-	114 Universal joints and drive shafts not as specified.	1378AS110	

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TABLE II. Examination schedule. (Continued)

First Article Test	Quality Conformance Test	Defect	Drawing Number	Requirement Paragraph
X	-	115 Axles not as specified.	64A98D281	
X	-	116 Engine mounts not as specified.	1378AS109	
X	-	117 Wheels and tires and tubes when applicable not as specified.	-	3.2.2.1
X	-	118 Steering not as specified.	1378AS108	
X	X	119 Service brakes not as specified	-	3.7.3.4.1
X	X	120 Parking brake not as specified.	-	3.7.3.4.2
X	-	121 Controls and instrumentation not as specified.	1378AS102	
X	-	122 Lifting and tiedown provisions not as specified.	1378AS238	
X	-	123 Electric system not as specified.	1378AS160	
X	X	124 Battery, battery mounting, and terminals not as specified.	-	3.7.3.2.1 3.7.3.2.1.1
X	X	125 Treatment and painting not as specified.	-	3.5.3
X	X	126 Identification markings not as specified.	-	3.10.2

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TABLE II. Examination schedule. (Continued)

First Article Test	Quality Conformance Test	Defect	Drawing Number	Requirement Paragraph
X	X	127 Instruction plates not as specified.	-	3.10.1
X	X	128 Reflective markings not as specified.	X	3.10.3
X	X	129 Workmanship not as specified.	-	3.11
X	-	130 Castings and forgings not as specified.	-	3.11.1
X	-	131 Metal fabrication not as specified.	-	3.11.2
X	-	132 Welding not as specified.	-	3.11.3
X	-	133 Bolted connections not as specified.	-	3.11.4
X	-	134 Machine work not as specified.	-	3.11.5
X	X	135 Instrument panel lights.	1378AS101	

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TEST METHOD NO. 1

SPEED

1. Test Course: Test course shall be a dry, clean, smooth, level surface of sufficient length to attain maximum speed prior to entering measured distance.

2. Test Apparatus:

a. Tape measure.

b. Stopwatch or electronic timer.

c. Tachometer.

3. Test Procedure:

d. Operate loader at governed speed for a 60-second duration.

b. Drive the loader sufficient distance to attain maximum speed prior to entering the measured test course.

c. Record length of measured distance and time to traverse this distance. The recording of engine speed at end of required distance for first produced loader test only. Measured distance shall not be less than 22 feet.

d. Perform steps b. and c. for a total of six runs, three times in each direction on the test track.

e. Travel speed shall be the average of the six runs.

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TEST METHOD NO. 2

TURNING RADIUS

1. Test Course: Dry, smooth, concrete marked with a curve centerline of 9 foot radius.

2. Test Apparatus:

   a. Tape measure.

3* Test Procedure:

   a. Drive loader without load on the curve centerline so that at full cramp angle the outer front wheel is tangent to the centerline.

   b. Drive loader forward. The curve projected by the outer rear wheel should be equal to 10 feet or less.

   c* Perform operation b. with loader turning to the left.

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TEST METHOD NO. 3

HIGH TEMPERATURE TEST

1. Test Apparatus:

a. Temperature chamber.

b. Thermocouples.

c. Battery hydrometer.

d. Recording instruments (temperature reading).

2. Test Condition: All references herein to +160°F shall be defined as +160°F, plus 0°F, minus 5°F.

3. Test Procedure:

a. Service the loader with applicable lubricants for +160°F operation including engine oil, transmission oil and drive train lubricants,

b. Check the specific gravity of each battery cell to insure full charge.

c. Subject the loader to steps (1) through (6) of MIL-STD-810, Method 501, Procedure II.

d. Subject the loader to an ambient temperature of +125°F until all parts and lube oil have stabilized at +125°F. Perform the test after stabilization is assured by temperature readings of +125°F from thermocouples located as follows:

(1) In the engine lubricating oil gallery and in the oil sump.

(2) In the hydraulic reservoir,

e. Maintain +125°F ambient temperature throughout the entire test.

f. One operator shall perform all priming and starting operations.

g. Crank the engine using the loader battery that is stabilized at +125°F, Substitute or supplementary batteries or other electrical cranking energy shall not be used.

h. Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained within 15 minutes after starting,

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TEST METHOD NO. 3 (Continued)

i. When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

(1) 10 minute idling.

(2) 10 minutes half throttle.

(3) 5 minutes full throttle.

Perform the 25 minute cycle three times.

j. Record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during engine warmup and cycling, including but not limited to the following:

(1) Engine oil.

(2) Hydraulic oil.

k. During engine cycling, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheeels.

(2) Operate, through four complete movement cycles, all brakes .

(3) Mount to and dismount from the operator's position on the loader.

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TEST METHOD NO. 4

LOW-TEMPERATURE TEST (-25°F)

1. Test Apparatus :

   a. Cold chamber.

   b. Arctic clothing.

   c. Battery hydrometer.

   d. Voltmeter.

2. Test Condition: All references herein to -25°F shall be defined as -25°F, plus 0°F, minus 5°F.

3. Test Procedure:

   a. Service the loader with applicable lubricants for -25°F operation including engine oil, transmission oil, and drive train lubricants.

   b. Check the specific gravity of each battery cell to insure full charge.

   c. Subject the loader to an ambient temperature of -25°F until al parts, lube oil, and coolant, nave stabilized at -25°F.

   d. One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.

   e. Crank the engine using the loader battery that is cold soaked and stabilized at -25°F. Substitute or supplementary batteries or other electrical cranking energy shall not be used.

   f. Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained with 15 minutes after starting.

   g. When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

      (1) 10 minute idling.

      (2) 10 minutes half throttle,

      (3) 5 minutes full throttle.

      Perform the 25 minute cycle three times.

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TEST METHOD NO. 4 (Continued)

h. During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels.

(2) Operate, through two complete movement cycles, all brakes.

(3) Mount to and discount from the operator's position on the loader.

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TEST METHOD NO. 5

HUMIDITY

1. Test Apparatus: Humidity-temperature chamber and associated equipment.

2. Test Procedure:

a. Subject the loader to the humidity test of MIL-STD-810, Method 507, Procedure V.

b. After completion of step 4 of a. above, start the loader and run the engine for 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

(1) 10 minute idling,

(2) 10 minutes half throttle.

(3) 5 minutes full throttle.

During engine cycling the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels.

(2) Operate, through two complete movement cycles, all brakes.

(3) Operate, through two complete cycles, all lights.

c. Repeat steps 4, 6, 7, and 8 of MIL-STD-810, Method 507, Procedure V MIL-STD-810 for a total of five cycles.

d. After completion of c. above, operate the loader as in b.

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-L-85532(AS)	Loader, Weapons, Air Launched, A/S32K-1D

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation.

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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